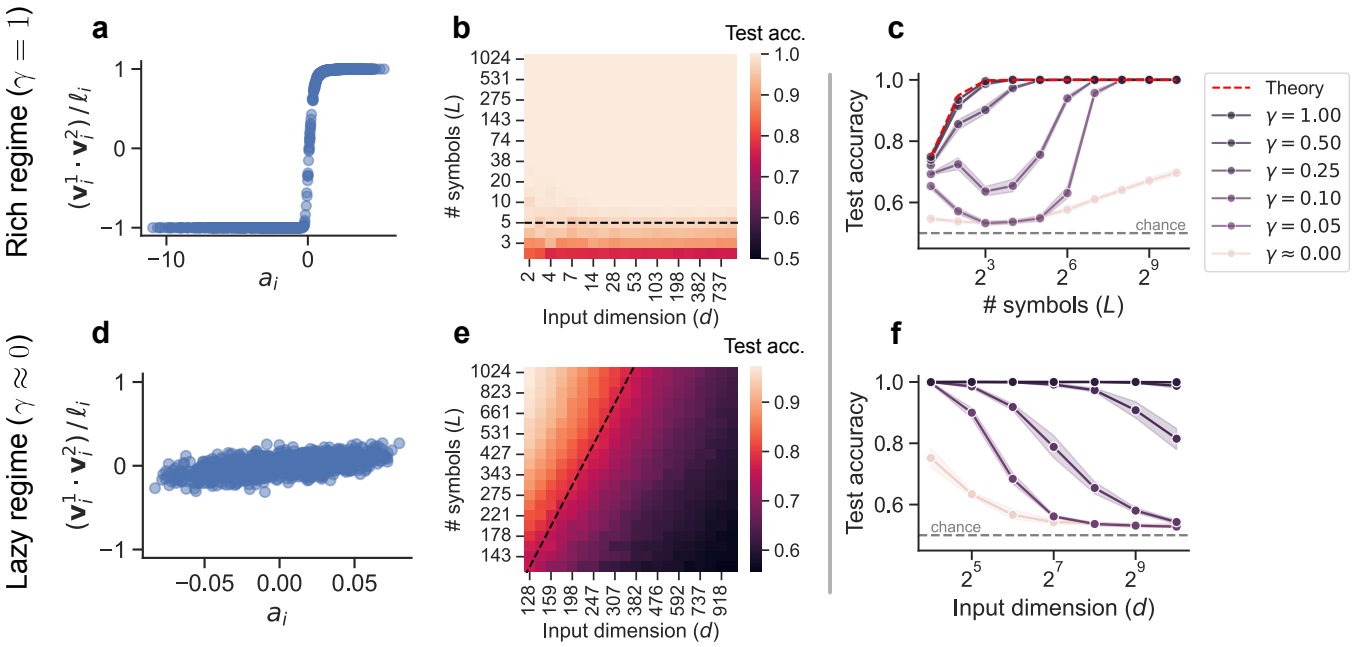

Figure 1: **Rich and lazy regime simulations.** We confirm our theoretical predictions with numeric simulations. **(a)**. Hidden weight alignment plotted against readout weights for a rich model. Weights become parallel or antiparallel, with generally higher magnitudes among negative readouts. **(b)** Test accuracy across different input dimensions and training symbols for a rich model ($m = 4096$). Accuracy is not affected by input dimension. **(c)** Test accuracy across different numbers of training symbols, for varying learning richness ($d = 256$, $m = 1024$). Richer models attain high performance with substantially fewer training symbols. The theoretically predicted rich test accuracy shows excellent agreement with our richest model. Finer-grain validation is plotted in Figure C3. **(d)** Hidden weight alignment plotted against readout weights for a lazy model. There is some correlation between alignment and readout weight, but the weights are nowhere near as close to being parallel/antiparallel as in the rich regime. **(e)** Test accuracy across input dimensions and training symbols for a lazy model ($m = 4096$). Accuracy is substantially affected by input dimension. Theory predicts that the number of training symbols required to maintain high accuracy scales at worst as $L \propto d^2$, plotted in black. **(f)** Test accuracy across different input dimensions, for varying learning richness ($L = 16, m = 1024$). Richer models show less performance decay with increasing dimension. **(all)** Results are computed across six runs. Shading corresponds to empirical 95 percent confidence intervals.

a rich-regime MLP attains the hand-crafted solution through training. It leverages its conceptual representations to learn the task with few training symbols and insensitivity to the input dimension (Section 3.2), exhibiting conceptual behavior. In contrast, a lazy-regime MLP is unable to adapt its representations to the task, and consequently incurs a high training cost and substantial sensitivity to input dimension (Section 3.3), exhibiting perceptual behavior. In an extension to a noisy version of our task, we show that a rich MLP approaches Bayes optimal performance under a generalizing prior across different noise variance σ^2 (Section 3.3). We validate our results on more complex, image-based tasks in Section 4, and discuss broader implications in Section 5.

3.1 Hand-crafted solution

To establish whether an MLP can solve the same-different task at all, we first outline a hand-crafted solution using $m = 4$ hidden units. Let $\mathbf{1} = (1, 1, \ldots, 1) \in \mathbb{R}^d$. Define the weight vector $\mathbf{w}_1^+$ by concatenation: $\mathbf{w}_1^+ = (\mathbf{1}; \mathbf{1}) \in \mathbb{R}^{2d}$. Further define $\mathbf{w}_2^+ = (-\mathbf{1}; -\mathbf{1})$, $\mathbf{w}_1^- = (\mathbf{1}; -\mathbf{1})$, and $\mathbf{w}_2^- = (-\mathbf{1}; \mathbf{1})$. Let $a^+ = 1$ and $a^- = \rho$, for some value $\rho > 0$. Our MLP is given by

$$f(\mathbf{x}) = a^+ \left(\phi(\mathbf{w}_1^+ \cdot \mathbf{x}) + \phi(\mathbf{w}_2^+ \cdot \mathbf{x})\right) - a^- \left(\phi(\mathbf{w}_1^- \cdot \mathbf{x}) + \phi(\mathbf{w}_2^- \cdot \mathbf{x})\right). \quad (2)$$

Note that the weight vectors $\mathbf{w}_1^+, \mathbf{w}_2^+$, which correspond to the positive readout a^+, are *parallel*: their components point in the same direction with the same magnitude. Meanwhile, weight vectors $\mathbf{w}_1^-, \mathbf{w}_2^-$ corresponding to the negative readout a^- are *antiparallel*: their components point in exact opposite directions with the same magnitude. Only the sign of f impacts the classification, so we assign $a^+ = 1$ and $a^- = \rho$ to represent the relative magnitude of a^- against a^+.

To see how this weight configuration solves the same-different task, suppose we receive a *same* example $\mathbf{x} = (\mathbf{z}, \mathbf{z})$. Plugging this into Eq (2) reveals that the negative terms vanish through our antiparallel weights, leaving $f(\mathbf{x}) = 2|\mathbf{1} \cdot \mathbf{z}| > 0$, correctly classifying this example.

Now suppose we receive a *different* example $\mathbf{x}' = (\mathbf{z}, \mathbf{z}')$. Recall that these quantities are sampled independently as

$\mathbf{z}, \mathbf{z}' \sim \mathcal{N}(\mathbf{0}, \mathbf{I}/d)$. As a result, we can no longer rely on a convenient cancellation. The quantity $\phi(\mathbf{w}_1^+ \cdot \mathbf{x}') + \phi(\mathbf{w}_2^+ \cdot \mathbf{x}')$ is equal in distribution to the quantity $\phi(\mathbf{w}_1^- \cdot \mathbf{x}') + \phi(\mathbf{w}_2^- \cdot \mathbf{x}')$, with respect to the randomness in $\mathbf{x}'$. Hence, to implement a consistent negative classification, we need to raise the relative magnitude ρ of our negative readout weight. Indeed, we calculate $p(f(\mathbf{x}') < 0) = \frac{2}{\pi}\tan^{-1}(\rho)$, which approaches 1 for $\rho \gg 1$.[2] Hence, by maintaining a large negative readout, we classify negative examples correctly with high probability. An illustration of this solution is provided in Figure B1.

An MLP need not implement this precise weight configuration to solve the SD task. Rather, our hand-crafted solution suggests two general conditions:

1. **Parallel/antiparallel weight vectors**. Weights associated with positive readouts must be parallel, and weight associated with negative readouts must be antiparallel. This allows us to classify any *same* example by canceling the contribution from negative readouts.

2. **Large negative readouts**. The cumulative magnitude of the negative readouts must be larger than that of the positive readouts. This allows us to classify any *different* example by raising the contribution from negative readouts.

Observe also that parallel and antiparallel weights are suggestive of conceptual representations for sameness and difference. Parallel weights contribute to a *same* classification, and exemplify the structure of a *same* example: the two components point in the same direction. Antiparallel weights contribute to a *different* classification, and exemplify the structure of a *different* example: the two components point as far apart as possible. We look for parallel/antiparallel weight vectors as evidence for conceptual representations of our SD task.

3.2 Rich regime

The rich learning regime is characterized by substantial weight changes throughout the course of training. For the MLP given in Eq (1), larger values of γ lead to rich learning behavior. We allow γ to vary between 0 and 1. The range $\gamma > 1$ is considered in Figure C3, where we see that no qualitative changes to our results occur for larger values of γ.

To study the rich regime, we take two approaches. First, recent theoretical work (Morwani et al., 2023; Wei et al., 2019; Chizat & Bach, 2020) suggest that MLPs trained in a rich learning regime on a classification task discover a max margin solution: the weights maximize the distance between training points of different classes. We derive the max margin weights for an MLP with quadratic activations in Theorem 1, finding that the max margin solution consists of parallel/antiparallel weight vectors, just as required from our hand-crafted solution. We defer the proof of this theorem to Appendix C.

Theorem 1. *Let $\mathcal{D} = \{\mathbf{x}_n, y_n\}_{n=1}^P$ be a training set consisting of P points sampled across L training symbols, as specified in Section 2. Let f be the MLP given by Eq 1, with two changes:*

1. *Fix the readouts $a_i = \pm 1$, where exactly $m/2$ readouts are positive and the remaining are negative.*

2. *Use quadratic activations $\phi(\cdot) = (\cdot)^2$.*

For weights $\theta = \{\mathbf{w}_i\}_{i=1}^m$, define the max margin set $\Delta(\theta)$ to be

$$\Delta(\theta) = \arg\max_{\theta} \frac{1}{P} \sum_{n=1}^{P} \left[(2y_n - 1) f(\mathbf{x}_n; \theta)\right],$$

subject to the norm constraints $||\mathbf{w}_i|| = 1$. If $P, L \to \infty$, then for any $\mathbf{w}_i = (\mathbf{v}_i^1; \mathbf{v}_i^2) \in \Delta(\theta)$ and $\ell_i = ||\mathbf{v}_i^1||\,||\mathbf{v}_i^2||$, we have that $\mathbf{v}_i^1 \cdot \mathbf{v}_i^2/\ell_i = 1$ if $a_i = 1$ and $\mathbf{v}_i^1 \cdot \mathbf{v}_i^2/\ell_i = -1$ if $a_i = -1$. Further, $||\mathbf{v}_i^1|| = ||\mathbf{v}_i^2||$.

However, the max margin result does not use ReLU MLPs, relies on fixed readouts a_i, and says nothing about learning efficiency or insensitivity to spurious perceptual details, two additional properties we require from a conceptual solution. To address these shortcomings, we extend the analysis by proposing a heuristic construction that approximates a rich ReLU MLP as an ensemble of independent Markov processes (Section C.2). Doing so enables a deeper characterization of rich learning dynamics, resulting in the following approximation of the test accuracy. Given an unseen test point $\mathbf{x}, y$, and prediction $\hat{y}$,

$$p(y = \hat{y}(\mathbf{x})) \approx \frac{1}{2} + \frac{1}{2}\Phi\left(\sqrt{\frac{2(L^2 - L)}{13(\pi - 2)}}\right), \tag{3}$$

where L is the number of training symbols and Φ is the CDF of a standard normal distribution.[3] This estimate suggests that the model attains over 95 percent test accuracy with as few as $L = 5$ training symbols, and test accuracy does not change with different d.

We confirm our theoretical predictions with simulations in Figure 1. At the end of training, the hidden weights indeed become parallel and antiparallel, with negative coefficients gaining larger magnitude (Figure 1a). Figures 1b and c show that the rich model learns the same-different task with substantially fewer training symbols than lazier models, and exhibits excellent agreement with our theoretical test accuracy prediction. As predicted, the rich model's performance does not vary with input dimension (Figure 1b).

Altogether, the rich model develops conceptual representations, learns the same-different task given only a small number of training symbols, and exhibits clear insensitivity to input dimension. In this way, it exhibits conceptual behavior on the same-different task.

3.3 Lazy regime

The lazy learning regime is characterized by vanishingly small change in the model's hidden representations after training. Smaller values of γ lead to lazy learning behavior. The limit $\gamma \to 0$ corresponds to the Neural Tangent Kernel (NTK)

[2] Full details are recorded in Appendix B.

[3] This estimate is for $L \geq 3$. For $L = 2$, $p(y = \hat{y}) = 3/4$. See Section C.5 for details.

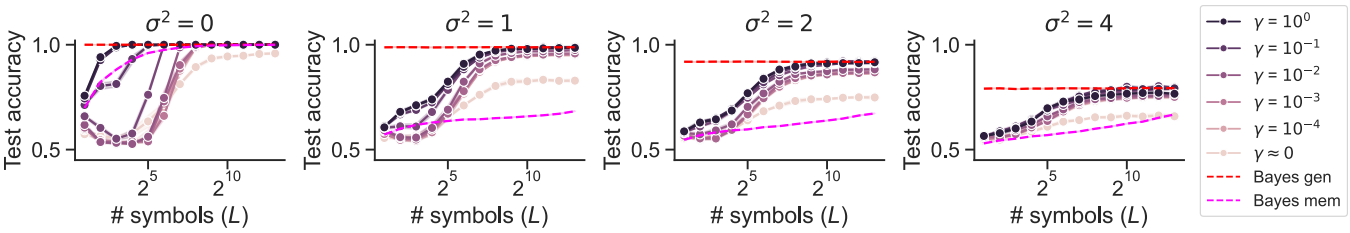

Figure 2: **Bayesian simulations.** Test accuracy across different numbers of training symbols, for varying richness and noise ($d = 64$, $m = 1024$). Bayes optimal accuracy for both generalizing and memorizing priors are plotted with dashed lines. In all cases, rich models attain the Bayes optimal test accuracy under a generalizing prior after sufficiently many training symbols. Shaded error regions are computed across six runs and correspond to empirical 95 percent confidence intervals.

regime, where the network is well-described by a linearization around its initialization (Jacot et al., 2018). In our numerics, we approximate this limit by using $\gamma = (1 \times 10^{-5})/\sqrt{d}$.

Because a lazy neural network cannot adapt its representations to an arbitrary pattern, it is impossible for a lazy MLP to learn parallel/antiparallel weights. However, because the statistics of a *same* example differ from that of a *different* example, it may still be possible for a lazy MLP to succeed at the task given enough training data[4]. Using standard kernel arguments (Cho & Saul, 2009; Jacot et al., 2018), we bound the test error of a lazy MLP in Theorem 2. The proof is deferred to Appendix D.

Theorem 2 (informal)**.** *Let f be an infinite-width ReLU MLP. If f is trained on a dataset consisting of P points constructed from L symbols with input dimension d, then the test error of f is upper bounded by $\mathcal{O}\left(\exp\left\{-L/d^2\right\}\right)$.*

This bound suggests that to maintain a consistently low test error (or equivalently, high test accuracy), the number of training symbols L needs to scale quadratically (at worst) with the input dimension: $L \propto d^2$.

We support our theoretical predictions with simulations in Figure 1. Because the model is in a lazy regime, the hidden weights do not move far from initialization, and no clear parallel/antiparallel structure emerges (Figure 1d). Figure 1c shows how models require increasingly more training data as richness decreases. Lazier models are also substantially more impacted by changes in input dimension (Figure 1e and f), and the scaling of training symbols with input dimension is consistent with our theory (Figure 1e).

Altogether, the lazy model is unable to learn conceptual representations, instead relying on statistical associations that require a large amount of training data to learn and exhibit strong sensitivity to input dimension. In this way, the lazy model exhibits perceptual behavior on the same-different task.

3.4 Same-different with noise

Up until now, we defined equality by exact identity: even a minuscule deviation in a single coordinate is enough to break equality and classify an example as *different*. Reality is far less clean, and real-world objects are rarely equal up to exact identity. As a first step towards this broader setting, we relax our dependence on exact identity and consider a noisy SD task. In the notation of our setup (Section 2), we allow $\sigma^2 > 0$.

To understand optimal performance under noise, we apply the following Bayesian framework. As a baseline, we consider a prior corresponding to an idealized model which *memorizes* the training symbols. This memorizing prior assumes every input symbol is distributed uniformly among the training symbols. To contrast this baseline, we consider a gold-standard prior corresponding to a model which *generalizes* to novel symbols. This generalizing prior assumes every input symbol follows the true underlying distribution. By comparing the test accuracy of the trained models to the posteriors computed in these two settings, we identify which prior more closely reflects the models' operation. The calculation of these posteriors are recorded in Appendix E.

Results are plotted in Figure 2. In all cases, we find that the rich model approaches Bayes optimal under the generalizing prior. Lazier models tend to plateau at lower test accuracies; they nonetheless tend to exceed the performance of the memorizing prior at higher noise, indicating some level of generalization. Overall, learning richness appears to support convincing generalization to novel training symbols in the noisy SD task.

4 Validation in vision tasks

To validate our theoretical findings in a more complex, naturalistic setting, we turn to visual same-different tasks. Specifically, we examine three datasets designed originally to study visual reasoning and computer vision: 1) PSVRT (Kim et al., 2018), 2) Pentomino (Gülçehre & Bengio, 2016), and 3) CIFAR-100 (Krizhevsky & Hinton, 2009). These tasks offer significantly more challenge over the simple SD task we examine before. Rather than reason over symbol embeddings, a model must now reason over complex visual objects. Inputs are now images, and equality is no longer exact identity: inputs can be equal up to translation (in PSVRT), rotation (in Pentomino), or merely share a class label (CIFAR-100). All additional details on model and task configurations are enumerated in Appendix G.

[4]For example, for a *same* input $\mathbf{x} = (\mathbf{z};\mathbf{z})$ and a *different* input $\mathbf{x}' = (\mathbf{z}_1, \mathbf{z}_2)$, the variance of $\mathbf{1} \cdot \mathbf{x}$ is twice that of $\mathbf{1} \cdot \mathbf{x}'$. Leveraging distinct statistics like this may still allow the lazy model to learn this task.